Multi-Scale Network Organization in AI-Assisted Knowledge Exploration

Alexander Towell

***Abstract*—As AI assistants become sustained intellectual partners, conversation archives accumulate latent cognitive structure invisible in sequential logs. We introduce a multi-scale network analysis framework (*cognitive MRI*) and apply it to 1,908 conversations spanning 29 months. Embedding each conversation as a vector and connecting those exceeding a cosine similarity threshold, we analyze the resulting network at three scales. Topologically, Louvain community detection reveals 15 knowledge communities with heterogeneous structure: hub-and-spoke theoretical domains and tree-like applied domains ($Q = 0.750$). Temporally, monthly cumulative snapshots show super-linear densification ($\gamma = 1.405$, $R^2 = 0.993$) and sub-linear preferential attachment ($\beta = 0.763$), matching growth laws of collective knowledge systems such as citation and patent networks. Semantically, LLM-extracted concepts form a co-occurrence network with small-world topology ($\sigma \approx 6.6$), matching benchmarks for human semantic memory, and vocabulary growth following Heaps' law ($\beta_H = 0.320$ vs. null 0.268, $p < 0.001$). These findings suggest that AI-assisted knowledge exploration produces cognitive artifacts with deep structural regularity amenable to principled analysis.**

***Index Terms*—Cognitive networks, semantic similarity, knowledge graphs, densification, small-world networks, AI-assisted learning, conversation analysis.**

I. Introduction

Millions of people now conduct sustained intellectual exploration through AI assistants. These conversations accumulate into rich but opaque archives: sequential logs that hide structural relationships between topics, the dynamics of knowledge growth, and the conceptual organization that emerges over time.

Existing work on AI conversations focuses predominantly on single-turn quality metrics [1], prompt engineering, or content moderation. Large-scale conversation corpora such as WildChat [2] and LMSYS-Chat-1M [3] have been collected and analyzed for safety and preference, but not as cognitive artifacts with recoverable structure. No framework treats multi-month conversation archives as networks amenable to the same quantitative analysis applied to citation networks, collaboration graphs, or human semantic memory.

We address this gap with *cognitive MRI*: a multi-scale network analysis framework that transforms conversation archives into semantic similarity networks and reveals three layers of organization:

1) **Topological:** Community structure distinguishing theoretical from applied knowledge domains, with heterogeneous internal topology and three distinct bridge conversation types [4].
2) **Temporal:** Growth laws quantitatively matching collective knowledge systems, including super-linear densification and sub-linear preferential attachment.
3) **Semantic:** A latent concept hierarchy with small-world topology matching human semantic memory benchmarks and sublinear vocabulary growth.

The central finding is that a single individual's AI-assisted knowledge exploration produces cognitive artifacts exhibiting structural regularities previously observed only in large-scale collective systems [5], [6]. This suggests scale-bridging organizational principles underlying both personal and collective knowledge production.

Our approach differs from existing conversation analysis in an important way. While corpora like WildChat and LMSYS-Chat-1M prioritize breadth (millions of conversations from many users), we prioritize *depth*: 1,908 conversations from a single user over 29 months. This longitudinal single-user design enables temporal analysis impossible with cross-sectional multi-user data, revealing how knowledge networks grow, densify, and stabilize over time. The trade-off is generalizability, which we address in the discussion.

The remainder of this paper is organized as follows. Section II reviews related work in conversation analysis, network science, and cognitive networks. Section III describes the dataset and network construction pipeline. Section IV presents the three-scale analysis. Section V discusses implications and limitations, and Section VI concludes.

II. Related Work

A. AI conversation corpora

Several large-scale AI conversation datasets have been released for research. WildChat [2] comprises one million ChatGPT interactions collected via a browser extension, enabling studies of user behavior, toxicity, and language distribution. LMSYS-Chat-1M [3] provides conversations across 25 language models, focusing on model comparison and preference learning. OpenAssistant [7] crowdsourced 161,443 messages in 35 languages for alignment research. These corpora are studied primarily for content analysis and model evaluation. None applies network-theoretic methods to uncover structural relationships between conversations or to model knowledge organization over time.

B. Network science of knowledge systems

Network science has a rich tradition of analyzing knowledge systems. Citation networks exhibit densification laws [5], preferential attachment [8], and community structure [9]. Co-authorship networks reveal the social structure of science [10],

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while science-of-science models track how disciplines emerge, grow, and interact [11]. De Solla Price [12] established that citation networks grow through cumulative advantage, a principle later formalized as preferential attachment by Barabási and Albert [8]. Leskovec et al. [5] showed that many real networks densify over time, with edges growing super-linearly relative to nodes.

Temporal network analysis [13] provides frameworks for studying how network structure evolves, including community tracking methods that detect births, deaths, merges, and splits [14], [15]. These tools have been applied extensively to social networks and collaboration graphs but not to personal AI conversation archives, which differ from traditional knowledge networks in being generated by a single individual interacting with an AI system rather than by a community of researchers.

C. Topic modeling and knowledge extraction

Traditional approaches to understanding document collections include Latent Dirichlet Allocation (LDA) [16], which models documents as mixtures of topics, and neural topic models such as BERTopic [17], which use pre-trained embeddings for topic discovery. These methods produce flat or hierarchical topic assignments but do not model structural relationships between documents.

More recently, Graph RAG [18] constructs entity-level knowledge graphs from document collections to improve retrieval-augmented generation. While Graph RAG builds inter-document structure, it targets information retrieval rather than cognitive analysis and does not model temporal evolution or growth laws. Our approach is complementary: cognitive MRI treats conversations as atomic units connected by semantic similarity, enabling network-level analysis of community structure, growth dynamics, and conceptual hierarchy that topic models and knowledge graphs do not capture.

D. Cognitive network science

Cognitive network science [19] applies graph-theoretic methods to model mental representations. Steyvers and Tenenbaum [6] demonstrated that human semantic networks (Roget's Thesaurus, WordNet, free word association) exhibit small-world topology with high clustering and short path lengths. Collins and Loftus [20] proposed spreading activation in semantic networks as a model of memory retrieval. The extended mind thesis [21] and distributed cognition [22] argue that cognitive processes extend beyond the brain into external artifacts, including written records and computational tools. Our work operationalizes this perspective: the conversation archive functions as an externalized cognitive workspace whose network structure reflects the user's evolving knowledge organization.

E. Comparison with existing approaches

Table I positions our framework against existing conversation analysis methods. Cognitive MRI is distinguished by its combination of temporal analysis, semantic hierarchy extraction, and network-theoretic methods applied to a longitudinal single-user archive.

III. Dataset and Network Construction

A. Dataset

The dataset comprises 1,908 ChatGPT conversations generated by a single user over 29 months (December 2022 to April 2025), spanning machine learning, statistics, philosophy, programming, and cryptography. The archive covers five model generations: GPT-3.5 (Dec 2022–Mar 2023), GPT-4 (Mar–Nov 2023), GPT-4 Turbo (Nov 2023–May 2024), GPT-4o (May–Sep 2024), and reasoning models (Sep 2024–Apr 2025). This longitudinal span provides a natural experiment in how knowledge networks evolve as the underlying AI capability changes.

B. Embedding and weighting

Each conversation is embedded using nomic-embed-text [23], a reproducible text embedder producing 768-dimensional L2-normalized vectors with an 8,192-token context window. For each conversation, user messages and AI responses are concatenated separately. The final embedding is a weighted average:

$$\mathbf{v} = \frac{\alpha \cdot \mathbf{v}_{\text{user}} + \mathbf{v}_{\text{AI}}}{\|\alpha \cdot \mathbf{v}_{\text{user}} + \mathbf{v}_{\text{AI}}\|} \tag{1}$$

where α controls the relative influence of user vs. AI content.

C. Ablation study

The weighting parameter α raises a methodological question: whose content determines the semantic identity of a conversation? User messages reflect intent and framing, while AI responses contain detailed explanations and generated content. A 63-configuration ablation study across 9 weight ratios ($\alpha \in \{100, 10, 4, 2, 1, 0.5, 0.25, 0.1, 0.01\}$) and 7 similarity thresholds ($\theta \in \{0.80, 0.825, 0.85, 0.875, 0.90, 0.925, 0.95\}$) established that $\alpha = 2$ (user messages weighted 2:1 over AI responses) maximizes modularity at $Q = 0.750$ while preserving natural topic structure [4].

The result is intuitive: user-heavy embeddings ($\alpha \geq 10$) create overly fragmented communities because users phrase similar questions in diverse ways, while AI-heavy embeddings ($\alpha \leq 0.25$) merge distinct domains because AI responses share common explanatory patterns. The 2:1 ratio captures the user's topical intent while retaining enough AI content to disambiguate conversations where the user prompt is terse but the topic is specific.

Importantly, the underlying topic distribution remains remarkably stable across weight ratios: ML/AI content constitutes 26–37% and general topics 47–55% regardless of α, indicating that the community structure reflects genuine thematic organization rather than weighting artifacts.

D. Threshold selection and phase transition

Conversations are connected when their cosine similarity exceeds θ. Table III shows how network properties vary across thresholds, revealing a phase transition at $\theta \approx 0.875$.

Below $\theta = 0.875$, the network is over-connected (89–94% giant component, modularity below 0.65); above it, the

TABLE I
COMPARISON OF CONVERSATION ANALYSIS APPROACHES. COGNITIVE MRI UNIQUELY COMBINES MULTI-MONTH TEMPORAL ANALYSIS, SEMANTIC HIERARCHY EXTRACTION, AND NETWORK-THEORETIC METHODS ON A SINGLE-USER ARCHIVE.

Approach	**Scale**	**Network analysis**	**Temporal dynamics**	**Semantic hierarchy**	**Single-user depth**	**Growth laws**	**Conversations**
WildChat [2]	1M users	—	—	—	—	—	1,000,000
LMSYS-Chat-1M [3]	Multi-user	—	—	—	—	—	1,000,000
OpenAssistant [7]	Crowdsourced	—	—	—	—	—	66,497
BERTopic [17]	Varies	—	✓	—	—	—	Varies
GraphRAG [18]	Document	✓	—	✓	—	—	N/A
LDA [16]	Varies	—	—	—	—	—	Varies
Cognitive MRI (ours)	**Single user**	✓	✓	✓	✓	✓	**1,908**

TABLE II
DATASET COMPOSITION BY MODEL ERA. THE ARCHIVE SPANS FIVE CHATGPT MODEL GENERATIONS WITH INCREASING CAPABILITY.

Era	**Dates**	**Conv.**	**Months**
GPT-3.5	Dec 2022–Mar 2023	127	4
GPT-4	Mar–Nov 2023	298	8
GPT-4 Turbo	Nov 2023–May 2024	412	7
GPT-4o	May–Sep 2024	389	5
Reasoning	Sep 2024–Apr 2025	682	5
Total		**1,908**	**29**

TABLE III
NETWORK PROPERTIES ACROSS SIMILARITY THRESHOLDS. A PHASE TRANSITION NEAR $\theta = 0.875$ SEPARATES OVER-CONNECTED FROM FRAGMENTED REGIMES. THE CHOSEN THRESHOLD $\theta = 0.9$ MAXIMIZES MODULARITY WHILE RETAINING A 75% GIANT COMPONENT.

θ	Nodes	Edges	GC%	Q	Comm.
0.850	1,210	15,523	93.6	0.546	13
0.875	934	5,675	89.2	0.648	15
0.900	**601**	**1,718**	**75.4**	**0.750**	**15**
0.925	284	375	20.8	0.513	5

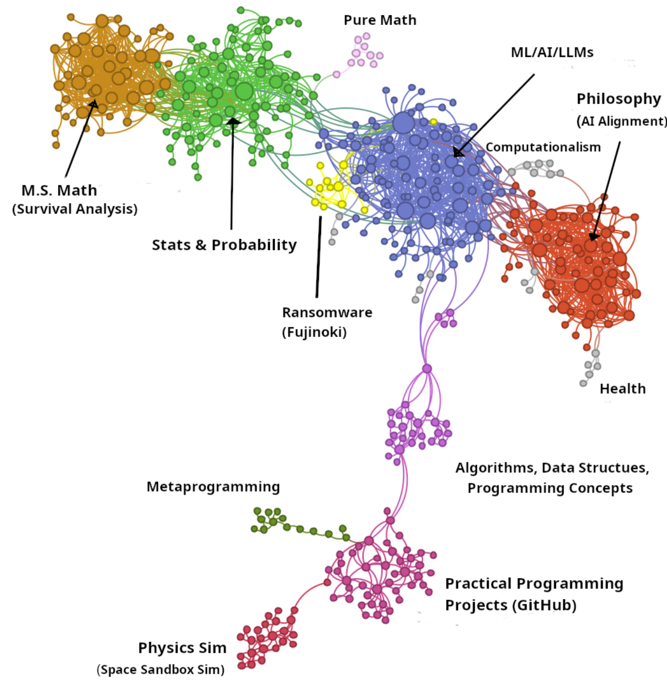

Fig. 1. Knowledge network at $\theta = 0.9$ showing 15 Louvain communities. Theoretical domains (ML/AI, Statistics, Philosophy, Mathematics) form dense hub-and-spoke structures; applied domains (Programming, Metaprogramming) form tree-like hierarchies. Node size is proportional to degree; layout by ForceAtlas2.

network catastrophically fragments (<21% giant component at $\theta = 0.925$). The chosen threshold of $\theta = 0.9$ lies in the post-transition regime where community structure is maximally resolved ($Q = 0.750$, 15 communities, 75.4% giant component). This 2.5% semantic distance between the over-connected and fragmented regimes represents a narrow window separating distinct cognitive contexts.

The phase transition itself is analytically interesting: it suggests that the semantic similarity landscape has a characteristic scale. Conversations cluster into groups separated by approximately 10% cosine distance ($1 - \theta = 0.1$), with much smaller within-group distances. This is consistent with the existence of genuine knowledge domains rather than a uniform continuum of topics.

IV. MULTI-SCALE ANALYSIS

A. Topological layer: knowledge communities

Louvain community detection [24] identifies 15 communities with modularity $Q = 0.750$ (Fig. 1). The network exhibits heterogeneous topology that distinguishes knowledge domain types.

TABLE IV
PROPERTIES OF THE FIVE LARGEST COMMUNITIES. THEORETICAL DOMAINS (ML/AI, STATS, PHILOSOPHY, MATH) SHOW HIGHER CLUSTERING AND CORE PERCENTAGES THAN APPLIED DOMAINS (PROGRAMMING).

Community	**Size**	$\bar{k}$	**Core%**	C	**Type**
ML/AI/LLM	103	8.60	31.1	0.465	Hub-spoke
Stats & Prob.	82	7.09	24.4	0.405	Hub-spoke
Philosophy	65	10.03	44.6	0.531	Hub-spoke
M.S. Math	44	13.45	59.1	0.576	Hub-spoke
Programming	45	3.78	8.9	0.396	Tree-like

Theoretical vs. applied structure. Table IV summarizes the five largest communities. Theoretical domains (ML/AI, Statistics, Philosophy, Mathematics) form dense hub-and-spoke structures with high clustering coefficients ($C > 0.40$) and substantial core membership (24–59%), suggesting that

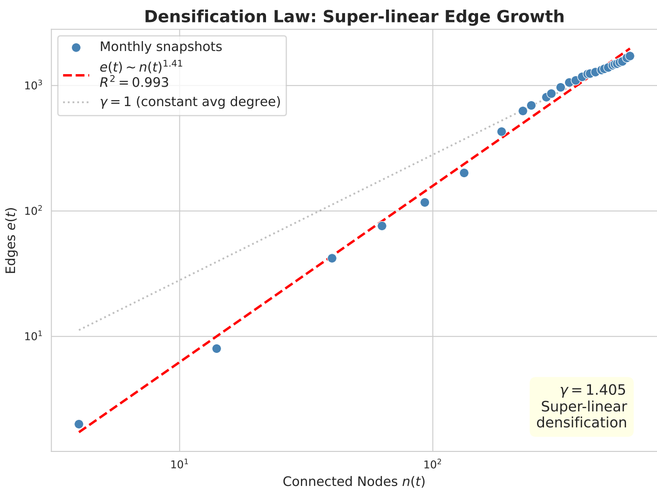

Fig. 2. Densification law: log-log plot of edges vs. connected nodes across 29 monthly snapshots. The power-law fit $e(t) \propto n(t)^{1.405}$ ($R^2 = 0.993$) indicates super-linear densification comparable to citation and patent networks [5].

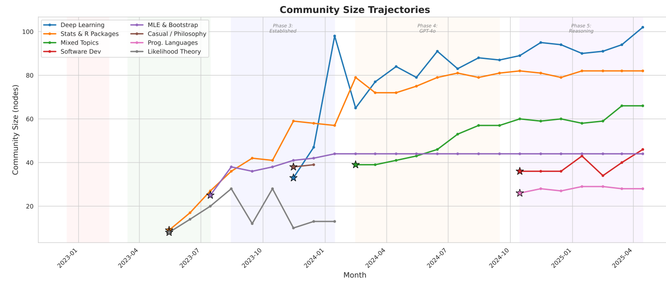

Fig. 3. Community size trajectories over 29 months. The two earliest communities (Statistics & R Packages, Deep Learning) exhibit first-mover advantage, remaining the largest throughout. Later communities (Software Development, Programming Languages) emerge during the reasoning model era. No merge or split events occur.

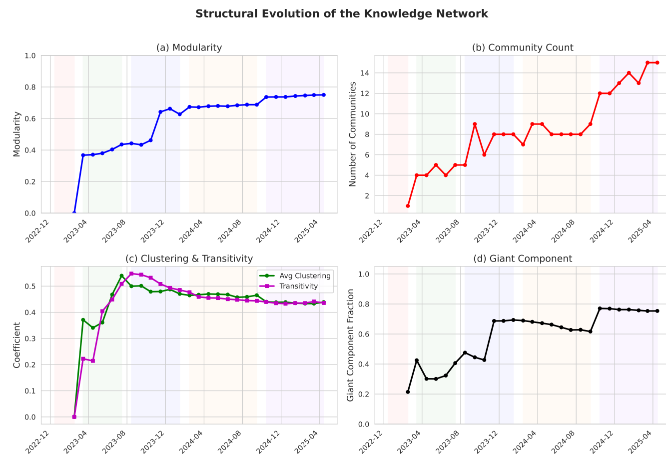

Fig. 4. Structural evolution over 29 months. (a) Modularity rises sharply in November 2023 and stabilizes above 0.7. (b) Community count grows from 1 to 15 with distinct plateaus. (c) Clustering coefficient and transitivity converge to 0.44. (d) Giant component fraction stabilizes at approximately 0.75. Vertical dashed lines indicate model-era transitions.

theoretical knowledge organizes around central concepts with many interconnections. The Mathematics community is particularly dense, with an average degree of 13.45 and 59.1% core membership, reflecting the tightly interconnected nature of mathematical concepts. Applied domains (Programming Projects, Metaprogramming) form tree-like hierarchies with lower density ($\bar{k} = 3.78$, core only 8.9%), reflecting the goal-oriented, sequential nature of practical work where each project branches from a parent concept without the dense cross-linking found in theoretical domains.

Core-periphery organization. A core-periphery decomposition [25] reveals that 25.6% of nodes constitute the core (average degree 18.94), while 74.4% form the periphery (average degree 3.15). Core conversations tend to address foundational concepts that connect multiple domains, while peripheral conversations are more specialized.

Bridge conversations. Three types of bridge conversations connect otherwise distant communities [4]: *evolutionary* bridges that accumulate connections over time as the user develops cross-domain understanding, *integrative* bridges that explicitly synthesize multiple domains in a single conversation, and *pure bridges* providing minimal but critical links between clusters. The highest-betweenness bridge (a conversation about geometric mean calculations) connects four communities with betweenness centrality of 45,468, illustrating how a single mathematical concept can serve as a cognitive linchpin across ML/AI, Statistics, Probability, and Tokenization.

This layer reveals *what* the user knows and how knowledge domains structurally relate.

B. Temporal layer: growth laws

Constructing 29 monthly cumulative snapshots reveals how the knowledge network develops over time (Figs. 2–4).

Super-linear densification. Edges grow faster than nodes following $e(t) \propto n(t)^{\gamma}$ with $\gamma = 1.405 \pm 0.022$ ($R^2 = 0.993$, $p < 10^{-29}$; Fig. 2). For every doubling of connected nodes, edges increase by a factor of $2^{1.405} \approx 2.65$. This places the network alongside citation networks (arXiv: $\gamma = 1.69$), patent networks ($\gamma = 1.26$), and autonomous systems ($\gamma = 1.18$) [5], indicating that individual knowledge exploration follows the same densification laws as collective knowledge production. The edges-per-node ratio increases from 0.5 (January 2023) to 2.86 (April 2025), confirming that the network fills in over time as new conversations connect to an increasingly dense existing structure.

Sub-linear preferential attachment. New edges attach to existing nodes with probability $\Pi(k) \propto k^{\beta}$ where $\beta = 0.763 \pm 0.074$ ($R^2 = 0.914$), measured via the kernel estimation method of Jeong et al. [26]. This lies between random attachment ($\beta = 0$) and the Barabási-Albert model ($\beta = 1$) [8], reflecting a balanced exploration-exploitation strategy [27]: popular topics attract further exploration, but less aggressively than in citation networks. The sub-linearity is statistically significant, with monthly z-scores against a random attachment null model consistently exceeding 8.4 during the exploration phase (March–July 2023).

Community stability. Modularity stabilizes at $Q > 0.6$ within 8 months and reaches 0.75 by the final snapshot. Tracking 40 communities across snapshots (Fig. 3) reveals 189

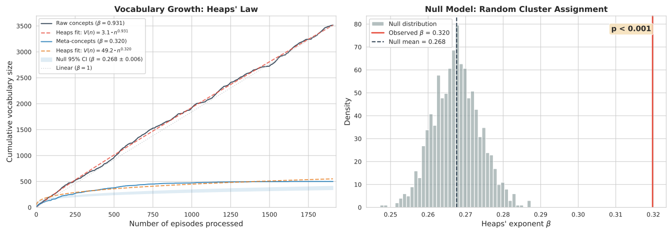

Fig. 5. Vocabulary growth of meta-concepts follows Heaps' law ($\beta_H = 0.320$), significantly exceeding the random clustering baseline ($\beta_{\text{null}} = 0.268$, $p < 0.001$). Left: cumulative growth curve with 95% null model band. Right: null distribution showing observed β_H above all random permutations.

continuations, 40 births, 25 deaths, and zero merge or split events, a pattern that contrasts sharply with social networks where communities routinely merge and split [15]. Knowledge domains grow internally rather than fusing. First-mover communities (Statistics & R Packages, Deep Learning) remain the largest throughout, exhibiting cumulative advantage analogous to de Solla Price's observation in citation networks [12].

Structural evolution across model eras. Fig. 4 shows that key structural metrics undergo a phase transition in November 2023, coinciding with the shift from GPT-4 to GPT-4 Turbo. Modularity jumps from 0.45 to 0.64 in a single month, the giant component fraction stabilizes, and the clustering coefficient converges to its final value of 0.44. This suggests that the network achieves structural maturity relatively early: after approximately 8 months and 400 conversations, the large-scale organization is established and subsequent growth fills in existing structure rather than reorganizing it.

The five model eras (Table II) contribute unevenly to network growth. The Reasoning era (Sep 2024–Apr 2025) accounts for 36% of all conversations (682 of 1,908) but produces proportionally fewer new communities, consistent with the pattern of internal growth rather than structural reorganization. Conversations from the GPT-3.5 era, despite being the smallest cohort (127 conversations), include several high-betweenness bridges, suggesting that early exploratory conversations play an outsized structural role.

This layer reveals *how* knowledge grows, and that individual development follows the same quantitative laws as collective systems.

C. Semantic layer: concept hierarchy

Moving beyond topological and temporal structure, we analyze the *conceptual content* of conversations using LLM-based concept extraction, connecting to the cognitive network science tradition of modeling mental representations as graphs [19].

Concept extraction pipeline. An LLM extracts 3–10 noun-phrase concepts per conversation (mean 3.96), yielding 6,275 raw concepts of which 3,517 are unique. Hierarchical clustering [28] on concept embeddings produces a four-level hierarchy: 500 meta-concepts (agglomerative clustering with Ward linkage), 50 themes, and 8 knowledge domains (Software Engineering 30.5%, Mathematics & Optimization 18%, Statistics 14.5%, Philosophy & AI Theory 13.9%, ML & Networks 7.8%, DevOps 7.8%, AI Safety 5%, LLM Engineering 2.4%). A bipartite episode-concept network (1,908 episodes, 500 meta-concepts, 7,153 edges) captures the many-to-many relationship between conversations and concepts.

Small-world topology. The concept co-occurrence network (499 nodes, 5,935 edges) exhibits small-world structure with $\sigma \approx 6.6$, computed against 100 Erdős-Rényi random graphs [29]: clustering is $6.89\times$ higher than random graphs ($C = 0.329$ vs. $C_{\text{rand}} \approx 0.048$) while path length is only $1.05\times$ longer ($L = 2.37$ vs. $L_{\text{rand}} \approx 2.25$). This is comparable to benchmarks for human semantic memory networks: Roget's Thesaurus ($\sigma \approx 13$), WordNet ($\sigma \approx 15$), and free word association norms ($\sigma \approx 5.6$) [6], [30], [31]. The result is robust across clustering granularities ($k = 100$–$1{,}000$), with $\sigma > 1$ throughout.

Sublinear vocabulary growth. Meta-concept vocabulary follows Heaps' law [32] with $\beta_H = 0.320$, significantly above a random clustering baseline ($\beta_{\text{null}} = 0.268$, $p < 0.001$; Fig. 5). Raw concept vocabulary grows nearly linearly ($\beta = 0.931$), confirming that the sublinear growth is a property of the semantic clustering rather than the raw data. Semantically coherent clusters create meaningful distinctions: new episodes require genuinely novel intellectual territory rather than recombining existing concepts.

Multi-domain integration. 77% of conversations span two or more knowledge domains, with 35% spanning three or more. Concept flow between domains is asymmetric: Software Engineering concepts appear in 40% of non-SE episodes (highest broadcast reach), while ML & Networks shows the highest porosity at 30% foreign concepts, consistent with its role as a methodological bridge domain. Size-normalized flow analysis reveals that domains are more semantically isolated than random (mean observed/expected $= 0.61$), but specific pairs show elevated cross-pollination: LLM Engineering concepts appear in ML & Networks episodes at $2.67\times$ the expected rate.

Orthogonality of topological and semantic structure. An important question is whether the concept-based domain structure merely recapitulates the topological communities found in Section IV-A. Normalized mutual information (NMI) between the 8 concept domains and the 15 Louvain communities is NMI $= 0.26$, indicating that these two decompositions capture substantially different aspects of the knowledge network. Topological communities are defined by pairwise conversation similarity (who links to whom), while concept domains are defined by shared intellectual content (what ideas recur together). The low NMI confirms that the semantic layer provides genuinely new information beyond what topology alone reveals: two conversations may share concepts without being topologically proximate, and vice versa.

This layer reveals *what concepts* organize knowledge and how they compose across domains. Table V synthesizes the three scales with literature comparisons.

V. Discussion

A. Cross-scale convergence

The three layers are not independent views but reveal a single organizing principle: AI-assisted exploration produces

TABLE V
MULTI-SCALE ORGANIZATION SUMMARY WITH LITERATURE COMPARISONS.

Scale	Key finding	Metric	Literature
Topological	15 communities; hub-spoke (theory) vs. tree-like (applied)	$Q = 0.750$; core 25.6%	Collab. networks
Temporal	Super-linear densification; sub-linear PA	$\gamma = 1.405$; $\beta_{\text{PA}} = 0.763$	arXiv γ=1.69, Patents γ=1.26
Semantic	Small-world concepts; Heaps' law; 77% multi-domain	$\sigma \approx 6.6$; $\beta_H = 0.320$	Roget's σ≈13, WordNet σ≈15

cognitive artifacts with structural regularities previously observed only in large-scale collective systems. A single individual's 29 months of AI conversation produces community structure comparable to scientific collaboration networks [9], [10], growth dynamics matching citation and patent networks [5], and semantic topology matching human semantic memory [6].

This convergence across scales suggests that the mechanisms driving knowledge organization may be scale-invariant: the same forces that shape how scientific communities collectively structure knowledge also shape how an individual organizes personal intellectual exploration. The sub-linear preferential attachment ($\beta = 0.763$) is particularly revealing: it indicates a balanced exploration-exploitation strategy where the user revisits established topics but maintains sufficient novelty-seeking to generate new communities. This balance, quantified through network growth dynamics, provides a formal measure of intellectual exploration style [27], [33].

B. Theoretical implications

The extended mind thesis [21] and distributed cognition [22] argue that cognitive processes extend into external artifacts. Our results provide quantitative evidence: the conversation archive functions as an externalized cognitive workspace whose network structure mirrors properties of internal semantic memory (small-world topology, sublinear vocabulary growth). This suggests that AI-assisted exploration does not merely retrieve information but actively co-constructs a structured knowledge representation whose organization follows universal principles.

The zero merge/split pattern in community evolution is notable. Unlike social networks where groups routinely merge and split [15], knowledge domains grow internally. This may reflect a fundamental difference between social and epistemic networks: knowledge domains have clearer boundaries defined by methodology and subject matter, whereas social groups are defined by fluid interpersonal connections.

C. The role of AI capability

The dataset spans five model generations with substantially different capabilities (Table II). Two observations suggest that network structure is driven primarily by the user's cognitive organization rather than by model properties. First, the densification exponent $\gamma = 1.405$ fits all 29 monthly snapshots with $R^2 = 0.993$, showing no discontinuity at model transitions. If model capability were driving network structure, we would expect the growth law to change when, for example, GPT-4's improved reasoning altered the nature of conversations. Second, the community structure that crystallizes by November 2023 (during the GPT-4 era) persists through GPT-4o and reasoning models without reorganization. New model capabilities produce new conversations, but those conversations attach to the existing community structure rather than reshaping it.

This stability is consistent with the interpretation that the network reflects the user's knowledge organization, which changes gradually through learning, rather than the AI's capabilities, which change abruptly through model updates. A formal test of this hypothesis would require comparing networks from the same user interacting with different models simultaneously, which is a direction for future work.

D. Practical implications

Conversation archives are underexploited as data sources. They contain recoverable cognitive structure applicable to several domains:

- **Personalized AI:** Network structure could inform recommendation of topics that bridge isolated knowledge communities.
- **Learning analytics:** Growth law parameters (γ, β) provide quantitative measures of learning trajectory, distinguishing broad explorers from deep specialists.
- **Knowledge auditing:** Community detection reveals blind spots and over-represented areas in a user's knowledge landscape.

The framework is applicable to any sustained human-AI interaction corpus, not only ChatGPT conversations.

E. Limitations and future work

This is a single-user case study ($n = 1$). The structural regularities may reflect individual cognitive style rather than universal laws; multi-user replication across diverse populations is the most important next step. LLM-based concept extraction introduces model-dependent bias: different extraction models may produce different hierarchies. Embedding-based similarity conflates topical with methodological similarity (two conversations about "optimization" in different domains may be connected despite serving different intellectual purposes).

The threshold selection at $\theta = 0.9$, while justified by the phase transition analysis, represents a single operating point. Future work should explore multi-resolution analysis across thresholds, potentially revealing fractal community structure.

Several additional directions merit investigation. First, causal analysis of whether network structure predicts learning outcomes: do users with higher densification exponents learn more effectively, or does a particular balance of exploration and exploitation correlate with knowledge retention? Second, real-time cognitive MRI as a feedback tool for AI-assisted learning, where a user could visualize their evolving knowledge network and receive recommendations for under-explored domains. Third, extension to multi-user collaborative archives, where network analysis may reveal collective knowledge dynamics in educational settings, research groups, or organizational knowledge management.

F. Reproducibility

All analysis code, embedding pipeline, and network construction scripts are available via the research compendium associated with the conference publication [4]. The conversation archive itself cannot be shared due to privacy considerations (it contains personal intellectual content), but the methodology is applicable to any conversation export in JSON format. The embedding model (nomic-embed-text) is open-source and deterministic given fixed inputs, ensuring reproducibility of the network construction step.

VI. Conclusion

We presented cognitive MRI, a multi-scale network analysis framework for AI conversation archives that reveals structure at three complementary scales. Applied to 1,908 conversations spanning 29 months and five model generations, the framework yields the following principal results:

1) **Topological structure:** 15 knowledge communities with modularity $Q = 0.750$, exhibiting heterogeneous topology (hub-spoke for theoretical domains, tree-like for applied domains) and a core-periphery organization where 25.6% of conversations form a densely connected core.
2) **Temporal growth laws:** Super-linear densification ($\gamma = 1.405$, $R^2 = 0.993$) and sub-linear preferential attachment ($\beta = 0.763$), placing the network alongside citation and patent networks in the landscape of real-world growth dynamics.
3) **Semantic hierarchy:** A four-level concept hierarchy (500 meta-concepts, 50 themes, 8 domains) with small-world topology ($\sigma \approx 6.6$) comparable to human semantic memory benchmarks, and vocabulary growth following Heaps' law ($\beta_H = 0.320$, $p < 0.001$ above random baseline).

The central finding is that individual-scale AI-assisted knowledge exploration exhibits quantitatively similar structural laws to collective knowledge systems. This convergence across scales, combined with the stability of network structure across model generations, suggests that the organizational principles reflect properties of human knowledge production rather than artifacts of any particular AI system. As AI-assisted exploration becomes ubiquitous, frameworks like cognitive MRI provide principled tools for understanding the cognitive structures that emerge from this new form of intellectual partnership.

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